

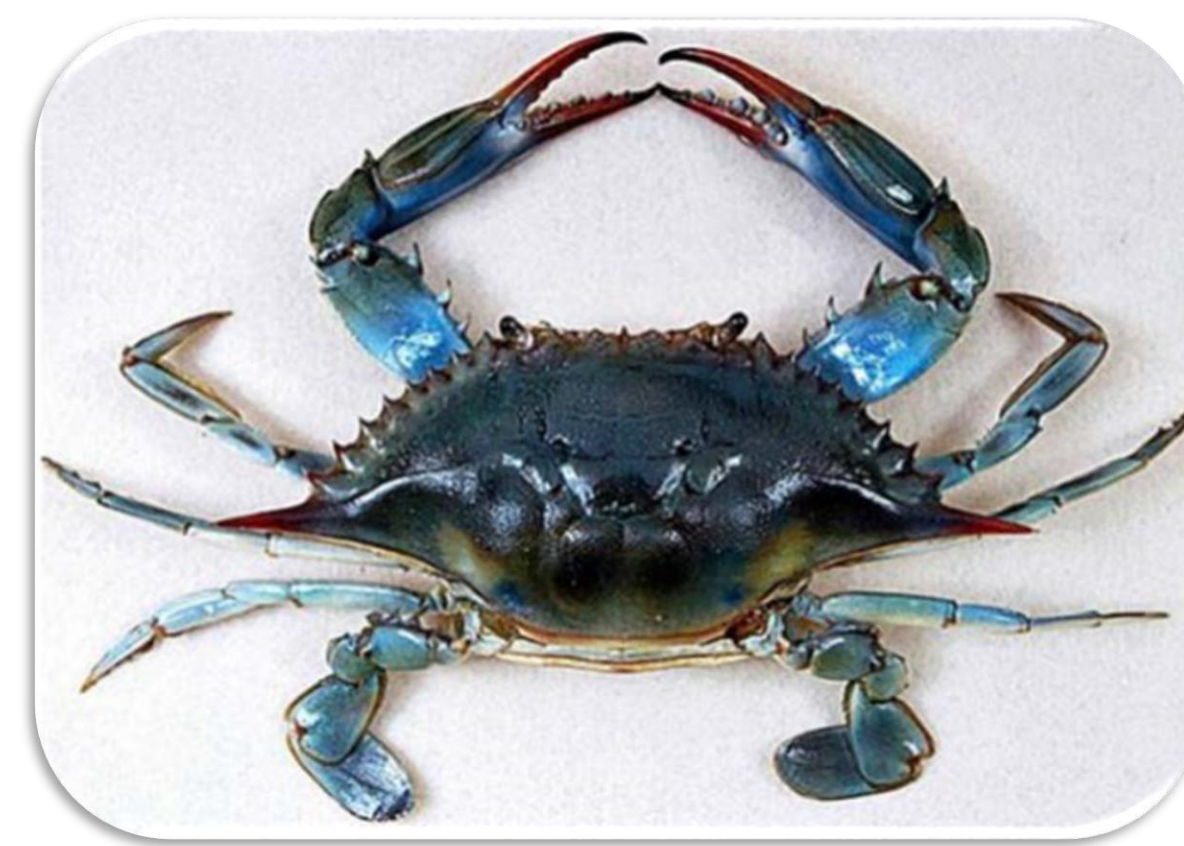
Beyond Traditional Test Species: Evaluating *Callinectes sapidus* as a Sentinel in Invaded Mediterranean Ecosystems

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Biomarker responses in blue crabs *C. sapidus* from two contrasting habitats are presented and compared with laboratory baseline levels to contextualize laboratory-derived thresholds within environmentally realistic exposure scenarios.

SEX	SITE	N	MEAN	SE	MIN	MAX
Total Length						
Females	Hondo	13	49.7	1.9	43.0	63.5
Males	Hondo	12	53.8	0.1	47.5	59.3
Females	Encañizadas	12	52.5	1.9	41.8	63.4
Males	Encañizadas	14	56.7	1.1	49.0	66.0
Females	Urrutias	0				
Males	Urrutias	20	66.4	1.5	57.7	80.0
Carapace width including spines						
Females	Hondo	13	118.2	6.3	97.1	161.0
Males	Hondo	12	123.2	2.3	108.5	135.0
Females	Encañizadas	12	111.4	6.4	87.4	152.8
Males	Encañizadas	14	117.3	3.4	99.1	143.3
Females	Urrutias	0				
Males	Urrutias	20	138.2	2.5	118.8	154.0
Total Weight						
Females	Hondo	13	85.6	9.5	51.0	155.0
Males	Hondo	12	104.7	5.3	74.4	134.6
Females	Encañizadas	12	97.5	10.6	55.1	178.9
Males	Encañizadas	14	132.2	11.3	71.4	223.0
Females	Urrutias	0				
Males	Urrutias	20	239.5	13.5	160.8	351.1



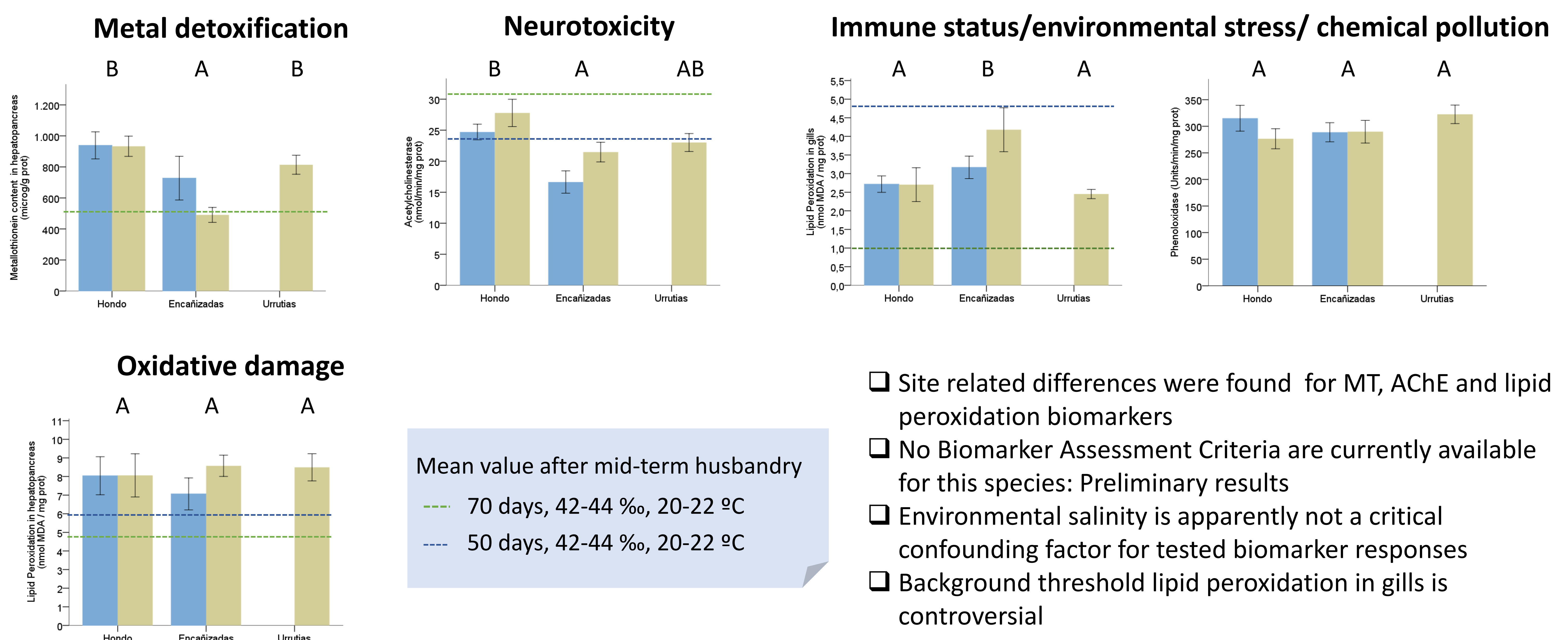
➤ Hondo Natural Park: Wetlands (high salinity gradient, 6 ‰)

➤ MM1: Encañizadas- Mediterranean influence (hypersaline, 40-47 ‰)

➤ MM2: Urrutias- Influenced by mining activity and agriculture runoff (Cartagena-La Unión)



SEX did NOT significantly influenced any of the analyzed biomarkers (t-test for the mean; p-value >0.05).



- ✓ Mar Menor crabs had approximately fivefold higher levels of Pb (0.9-1.0 mg/kg w.w.) than those from El Hondo Natural Park
- ✓ Cd concentration was double in crabs from MM2 (0.16 mg/kg w.w.) compared with crabs from other sites
- ✓ MT concentrations in crabs from all sites correlated significantly with Zn ($r > 0.7-0.8$) and Cd ($r > 0.6-0.8$) but not for Pb